Field notes from a river island.

Measuring water, soil and air on Cồn Hô, Vĩnh Long.

Aerial view of Cồn Hô

A forecast covers a grid. A farm is one plot.

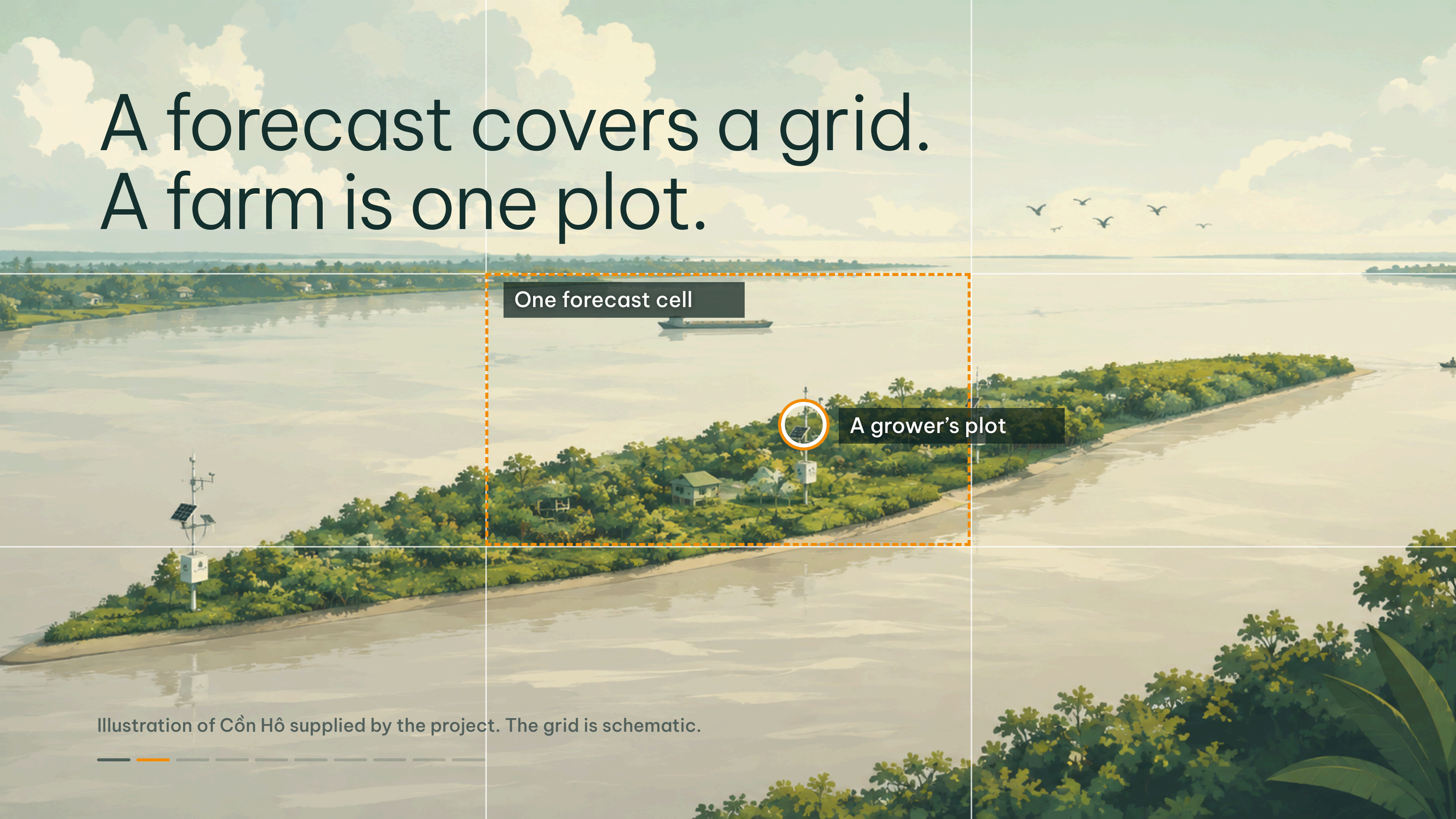


Illustration of Cồn Hô supplied by the project. The grid is schematic.

Measure.

Water, soil and air

Send.

By radio, then mobile data

Keep.

With its time and source

Show.

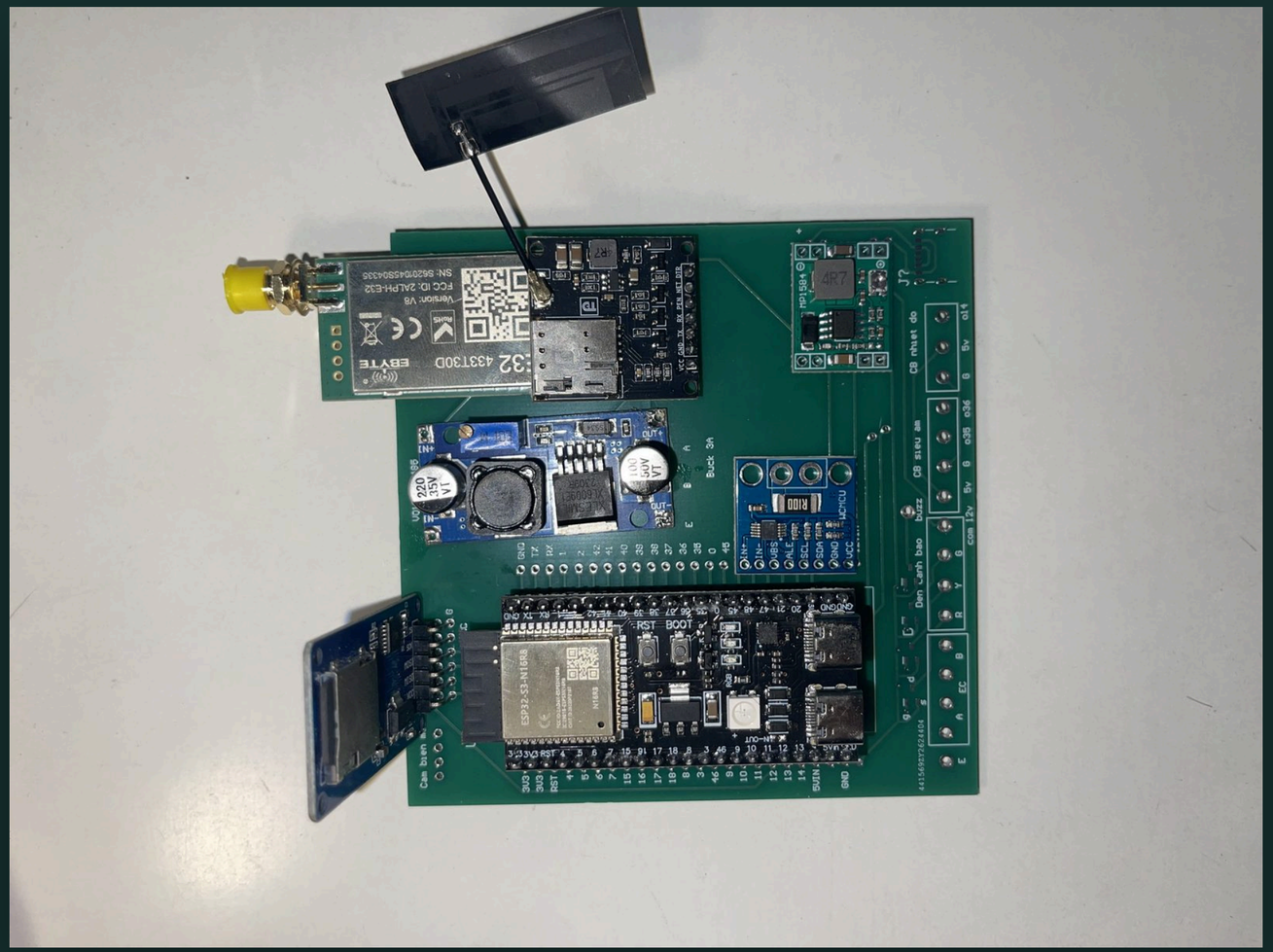
In public, with limits stated

Three nodes. One island.

A water station, a soil station, and a gateway that carries their data off the island.



Drawn by the project. Not a satellite image.



The gateway board we built. Bench photo.

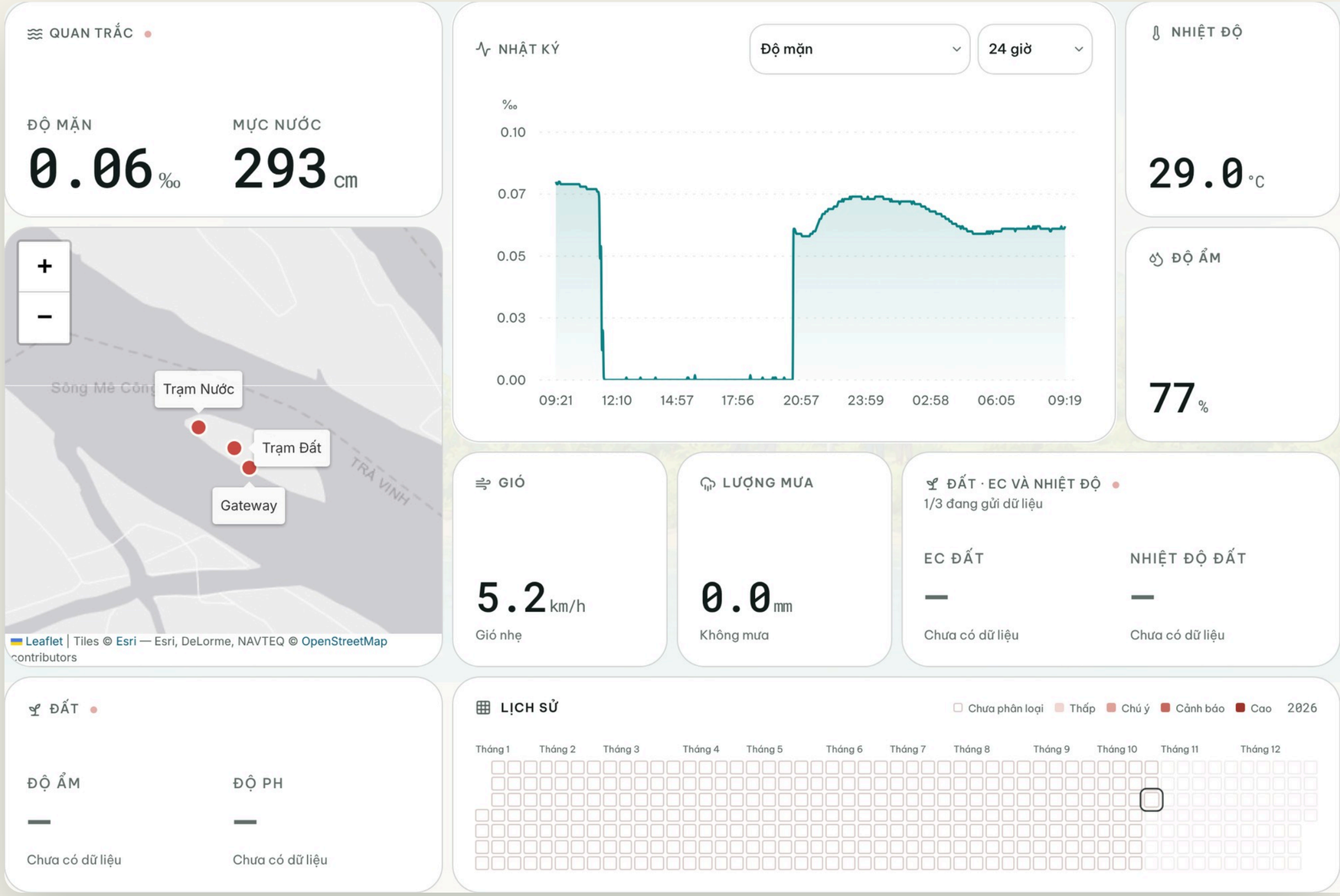
From the river to the screen.

Each hop keeps the time and the source of the reading.

Probe — Station — Gateway — Cloud — Observatory

What anyone can see today.

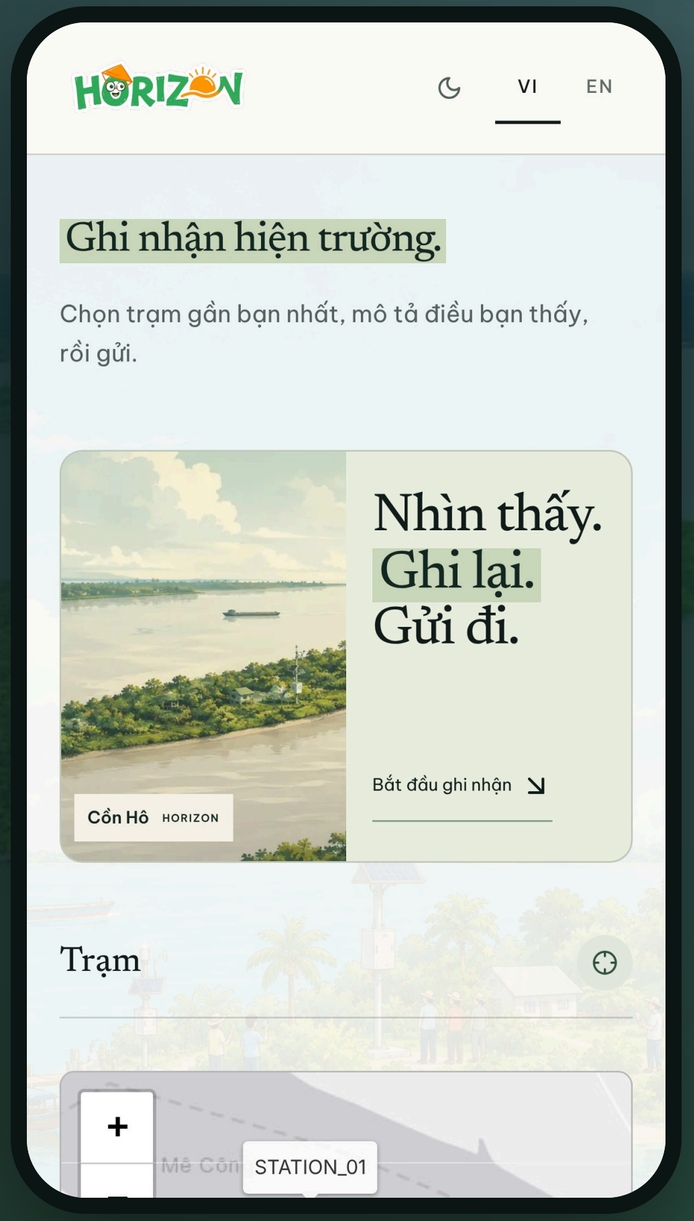
No login. Missing data stays missing, never zero.



Public Observatory, live capture, 21 Oct 2026.

Residents add what sensors cannot see.

Pick the nearest station, describe what you saw, add a photo or a short video. No account.



The report page on a phone.



Water conductivity probe. Bench photo, not an installed unit.

A working pilot, not yet a proven instrument.

Working

Readings travel from the island to a public page.

Still to prove

Calibration, field validation and long-term reliability.

From pilot to instrument.

Who operates it long term, and the yearly cost: [to confirm]

Harden

Close the open release checks.

[time to confirm]

Validate

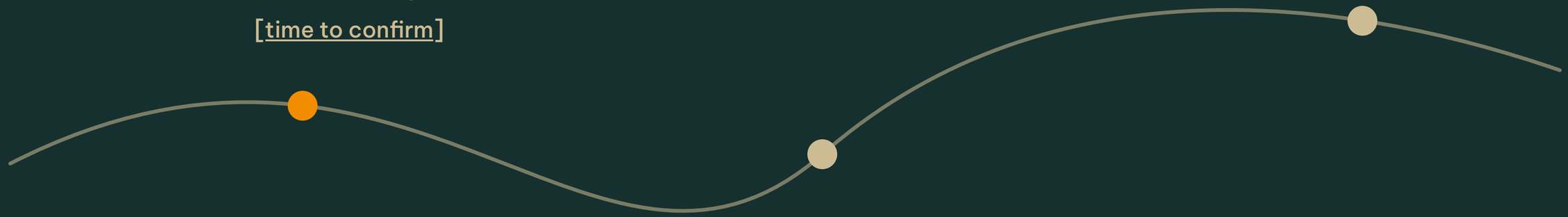
Calibrate on site, with expert review.

[time to confirm]

Use

Support irrigation and crop decisions.

[time to confirm]



Help turn a pilot into an instrument.

Funding

[Amount and period] for calibration and upkeep.

Field partners

[Growers and cooperatives] to use the readings.

Science advisors

[Agronomy and hydrology advisors] to validate.

[Name] · [email] · horizon.frogsleap.com.vn



Appendix

What has been checked

356 automated tests pass	Software tests of the web app and data handling, run on 21 Oct 2026. A software result, not a field result.
19 browser checks pass	Public flows tested in a browser, with a simulated storage service for uploads.
6,465 readings stored	From the water station since 25 Sep 2026, counted read-only on 21 Oct 2026. Shows data arrives, not that it is accurate.
Readings about every 2.5 minutes	Observed on the water station’s recent rows. Not a recorded network trace of the gateway link.

Appendix

Open findings from our audit

Zero-salinity runs	322 of the last 1,000 water-station readings store salinity as exactly 0 while flagged healthy. Cause not yet established; the probe may have been out of the water, or the register may read 0 at very low conductivity.
Soil station has no stored readings	The soil station has not yet stored any readings, so the soil section shows no data.
Water EC and temperature missing publicly	The public view shows salinity and level, but EC and temperature are empty. A database fix is written and reviewed, not applied.
Firmware fallback	If the salinity register fails, firmware estimates salinity from EC. It is an approximation sent as a reading. Recommended: send “missing” instead. Not changed; needs flashing approval.

Appendix

Before this is called production-ready

Hardware validation	A 20-step bench and supervised field gate is defined; no completed run is on record.
Credential rotation	Device and service credentials must be rotated before wider use.
Real-storage video upload	Direct upload is tested against a simulated service; a safe staging test is pending.
Network trace	A recorded trace of the gateway's DNS, TLS and upload path is pending.
Calibration and thresholds	Calibration against references and site-validated thresholds. Until then the Observatory states that no threshold is set.